

Nidhi Prashar

Chemistry



- 16+ Years of teaching experience
- Gold Medallist in the Master's Degree (Chemistry), UGC NET Qualified
- IJSO EXPERT with 150+ NSEJS Selections
- Mentored Science Olympiads GOLD MEDALISTS
- Teaching and mentoring students IIT, NEET, NSEC, IJSO exams
- Produced Single and double digit ranks in JEE/NEET (AIR 4 in AIIMS)

Examination Schedule

NSEA (Astronomy)

Saturday 21.11.26

2.30 pm to 4.30 pm

NSEP (Physics)

Sunday 22.11.26

8.30 am to 10.30 am

NSEC (Chemistry)

Sunday 22.11.26

11.30 am to 1.30 pm

NSEB (Biology)

Sunday 22.11.26

2.30 pm to 4.30 pm

NSEJS (Junior Science)

Sunday 22.11.26

2.30 pm to 4.30 pm

Telegram Channel



Grade : 8-10

Start Date: 26 April 2026

ALL INDIA NSEJS MOCK TEST SERIES-2026

✦✦ 7 Part Tests as per latest NSEJS Pattern

✦✦ 7 Full syllabus Tests as per
latest NSEJS Pattern

✦✦ Guidance on the basis
of Test Performance

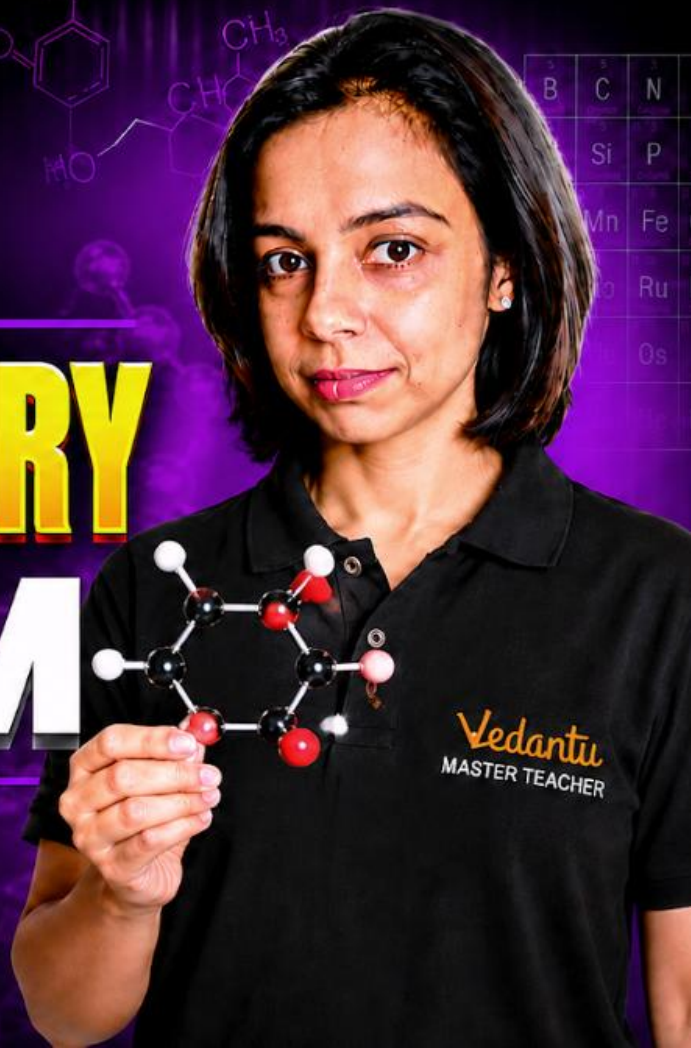
✦✦ Solution Videos

~~₹555~~ **₹499**

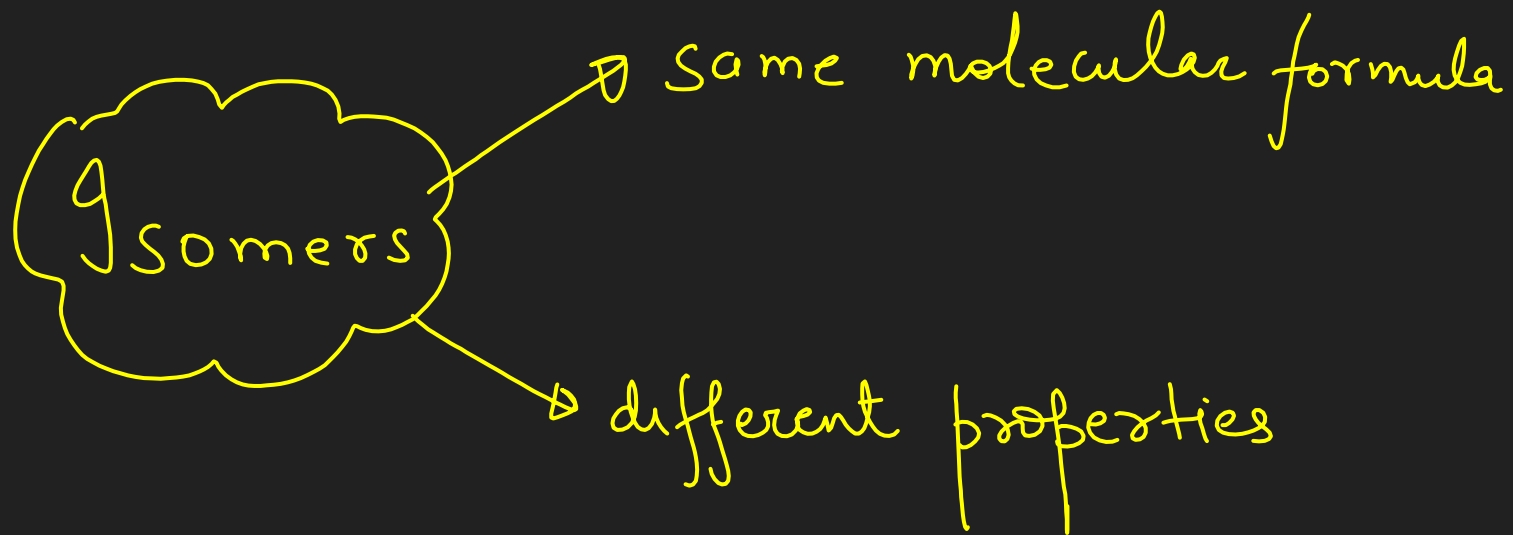


NSEJS CHEMISTRY

ORGANIC CHEMISTRY
ISOMERISM
ONESHOT



ISOMERISM



Introduction of Isomerism

Structural isomerism

Constitutional

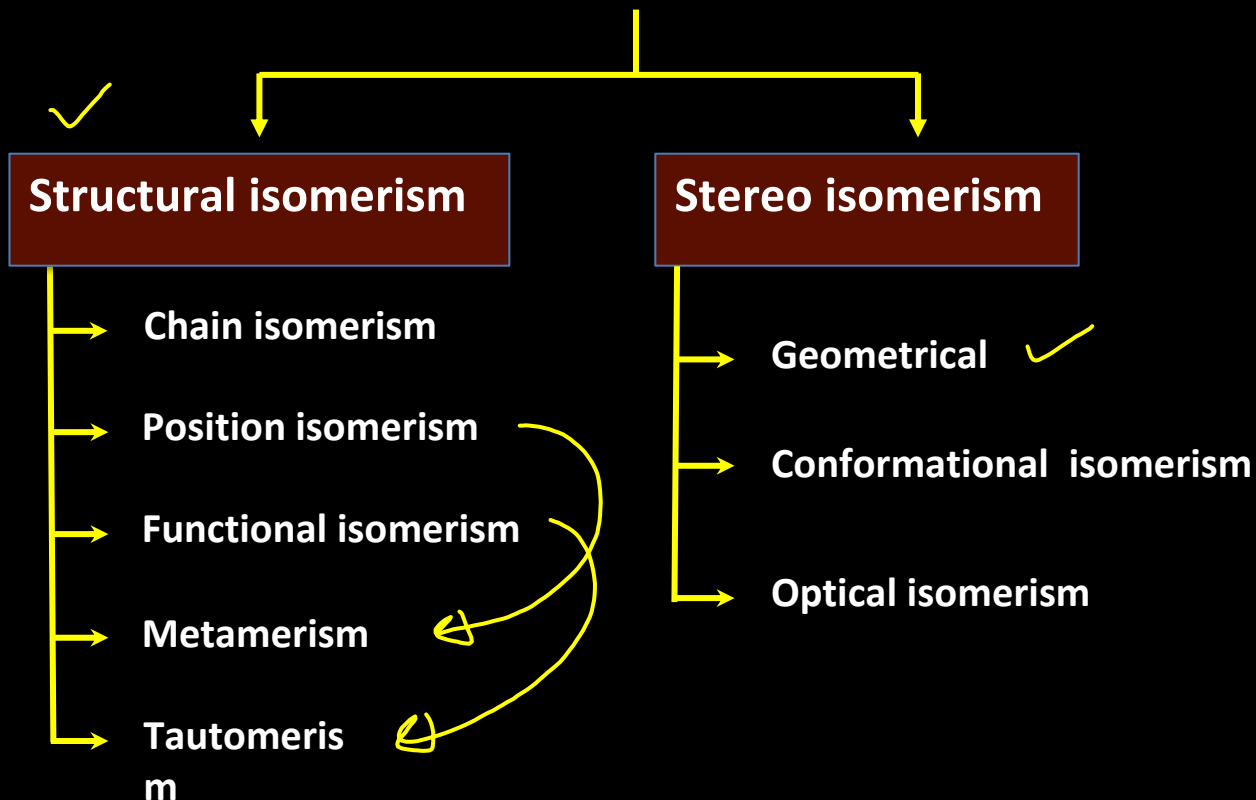
When the compounds have same molecular formula but different physical and chemical properties due to different connectivity (different constitution) of atoms then they are said to be Structural Isomers.

Stereo isomerism

Configurational

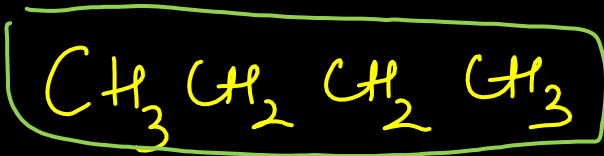
Compounds having same molecular formula but different arrangement of atoms in space are called stereoisomers and phenomenon is called as Stereoisomerism.

Types of isomerism



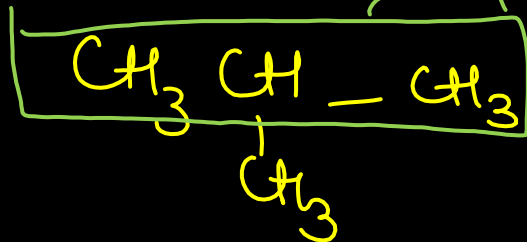
Structural Isomers

Butane



C_4H_{10}

propane



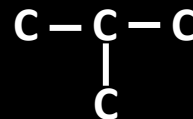
□ Chain isomerism:

It arises due to difference in the arrangement of the carbon atoms constituting the chain.

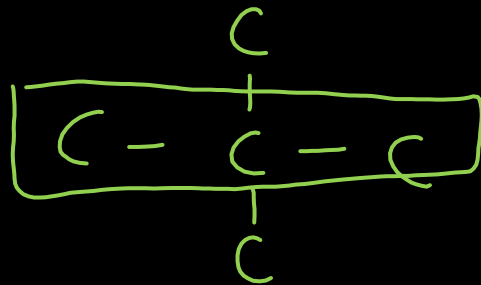
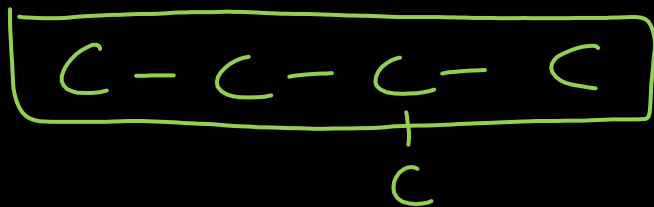
For example: Butane and 2-methylpropane are chain isomers.



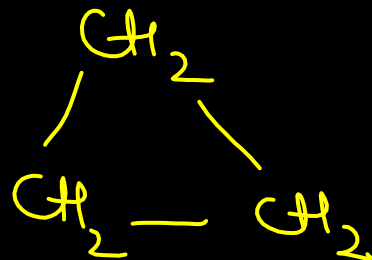
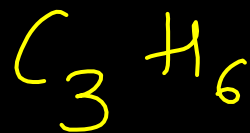
butane 4C



2 - methyl propane 3C



Ring Chain
Isomerism



Unsaturation Number

The presence of rings or double bonds within a molecule is indicated by a quantity called the unsaturation number of a hydrocarbon.

$$U = (C + 1) - \left(\frac{H - N}{2} \right)$$

Where: C → Total number of tetravalent atoms

H → Total number of monovalent atoms

N → Total number of trivalent atoms

Calculate the unsaturation number of the following compound $\text{C}_5\text{H}_6\text{Br}_2$

A

1

B

2

C

3

D

4

Calculate the unsaturation number of the following compound $\text{C}_5\text{H}_6\text{Br}_2$

A

1

B

2

C

3

D

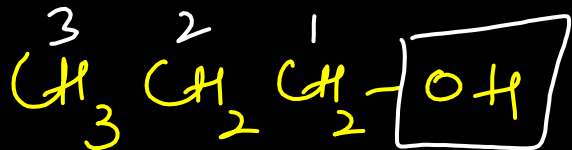
4

Solution :

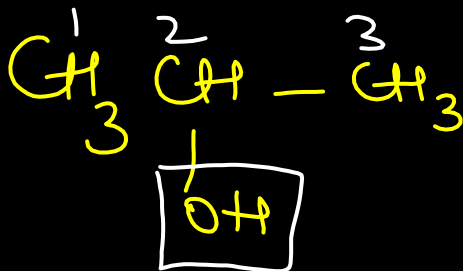
$$U = (5 + 1) - \left(\frac{6 + 2}{2} \right) = 6 - 4$$

$$U = 2$$

Position Isomers



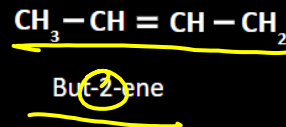
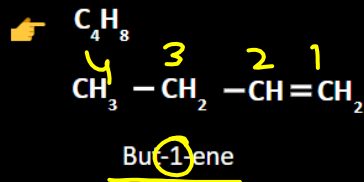
Propan-1-ol

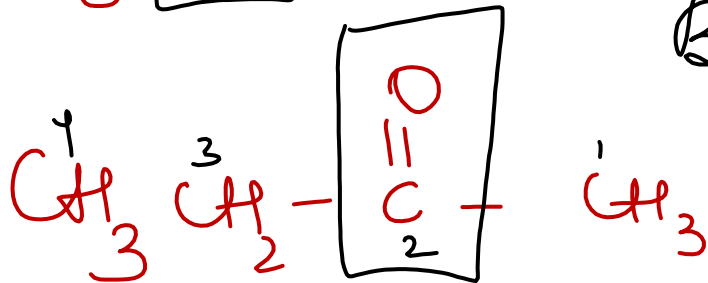
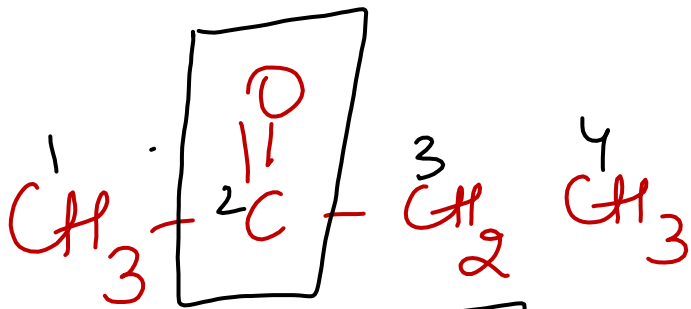


Propan-2-ol

Compounds having the same molecular formula but different in position substituents, $\text{C}=\text{C}$, $\text{C}\equiv\text{C}$ or functional group are called position isomers.

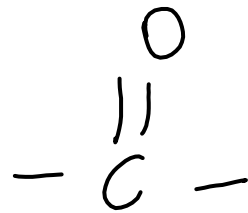
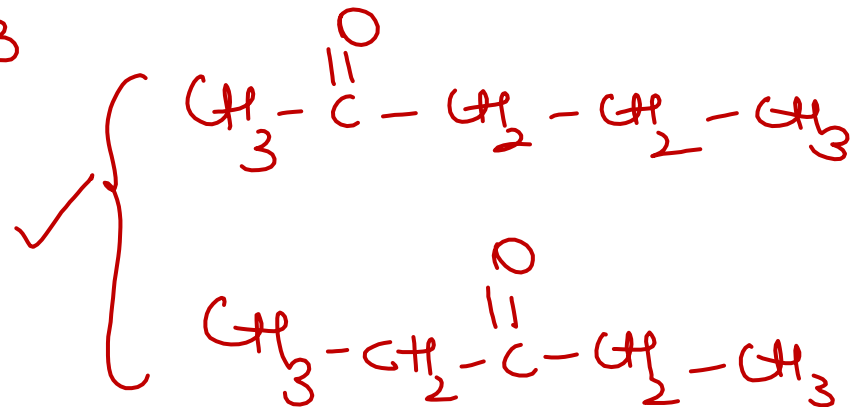
Example:



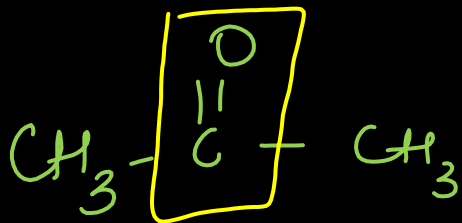


Butan-2-one Ketone

Butan-2-one



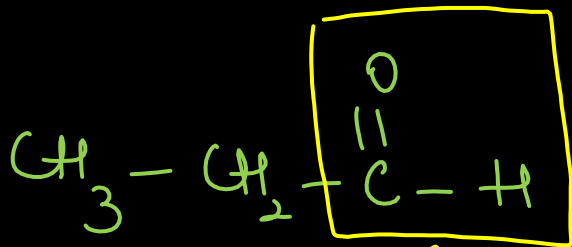
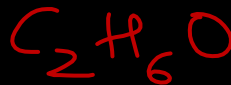
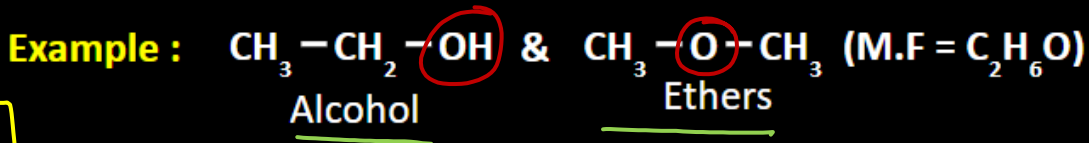
Functional isomers



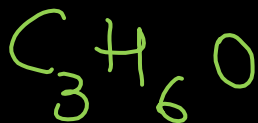
Ketone

It is observed due to the presence of different functional groups in a molecule

Example :



aldehyde



V. Imp

Functional isomers :

Homologous series formula	Family of compound
C_nH_{2n+2}	Alkane
C_nH_{2n}	Alkene, Cycloalkane
C_nH_{2n-2}	Alkyne, Cycloalkene, diene
$C_nH_{2n+2} \cdot O$ 	Alcohol & ether
<u>$C_nH_{2n}O$</u>	Aldehyde, ketone, unsaturated alcohols and cyclic ethers
$C_nH_{2n}O_2$	Carboxylic acids, Ester, hydroxy aldehyde or ketones

The number of structural isomers for C_6H_{14} is:

A

4

B

3

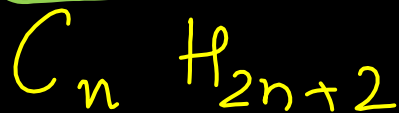
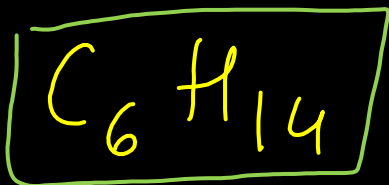
C

6

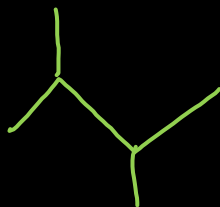
D

5

The number of structural isomers for C_6H_{14} is:



alkane



A

4

B

3

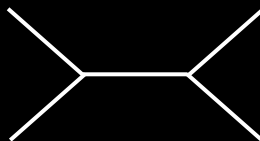
C

6

D

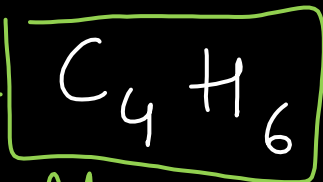
5

Solution :

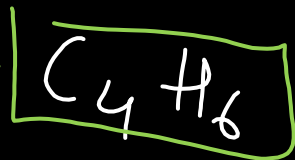


Functional isomer of but-1-yne is

alkyne



alkene



A

But-2-yne

B

But-1-ene

C

But-2-ene

✓
D

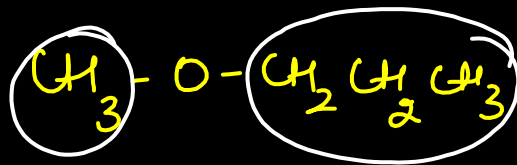
Buta-1,3-diene

Functional isomer of but-1-yne is

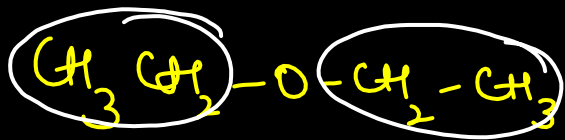
- A But-2-yne**
- B But-1-ene**
- C But-2-ene**
- D Buta-1, 3-diene**

Metamerism

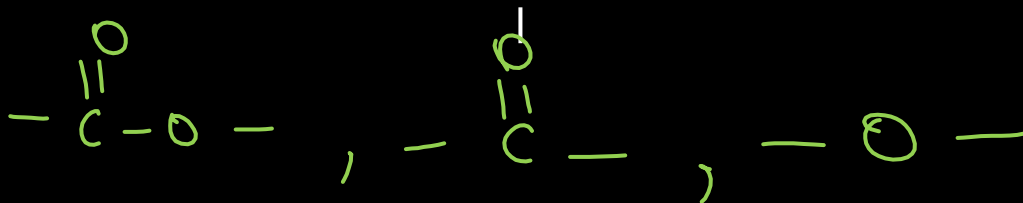
→ Position Isomerism



The Compounds having same molecular formula but different structural formulae due to different nature of alkyl groups present on either side of the polyvalent functional group are called Metamers and the phenomenon is known as Metamerism.

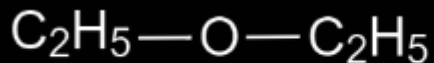
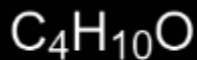


Polyvalent functional groups which shows metamerism are $-\text{O}-$, $-\text{S}-$, $-\text{NH}-$, $-\text{N}-$, and $-\text{CO}-$ etc.

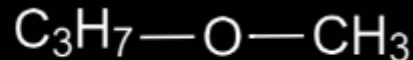


METAMERISM

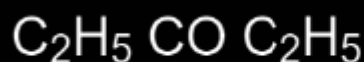
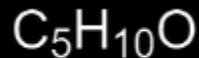
Examples:



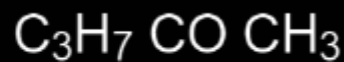
Diethyl ether



Methyl propyl ether



Diethyl Ketone



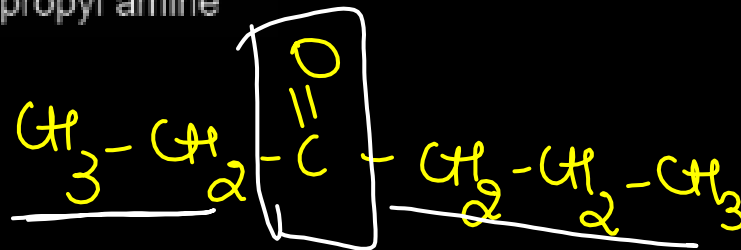
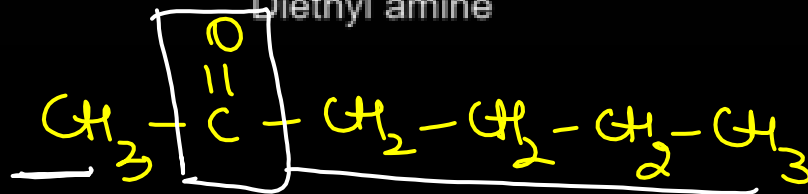
Methyl propyl Ketone

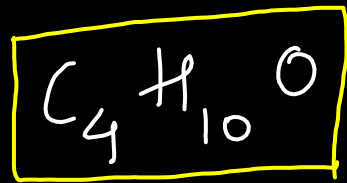


Diethyl amine



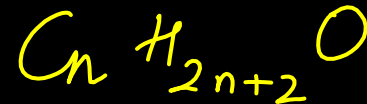
Methyl propyl amine





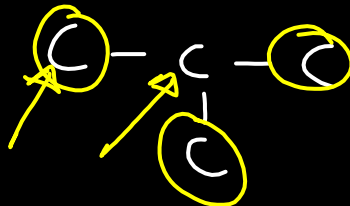
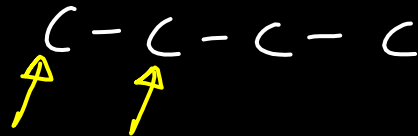
Total structural
Isomers

- Chain ✓
- Position ✓
- Functional ✓

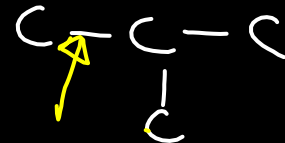


Alcohol

Ethers



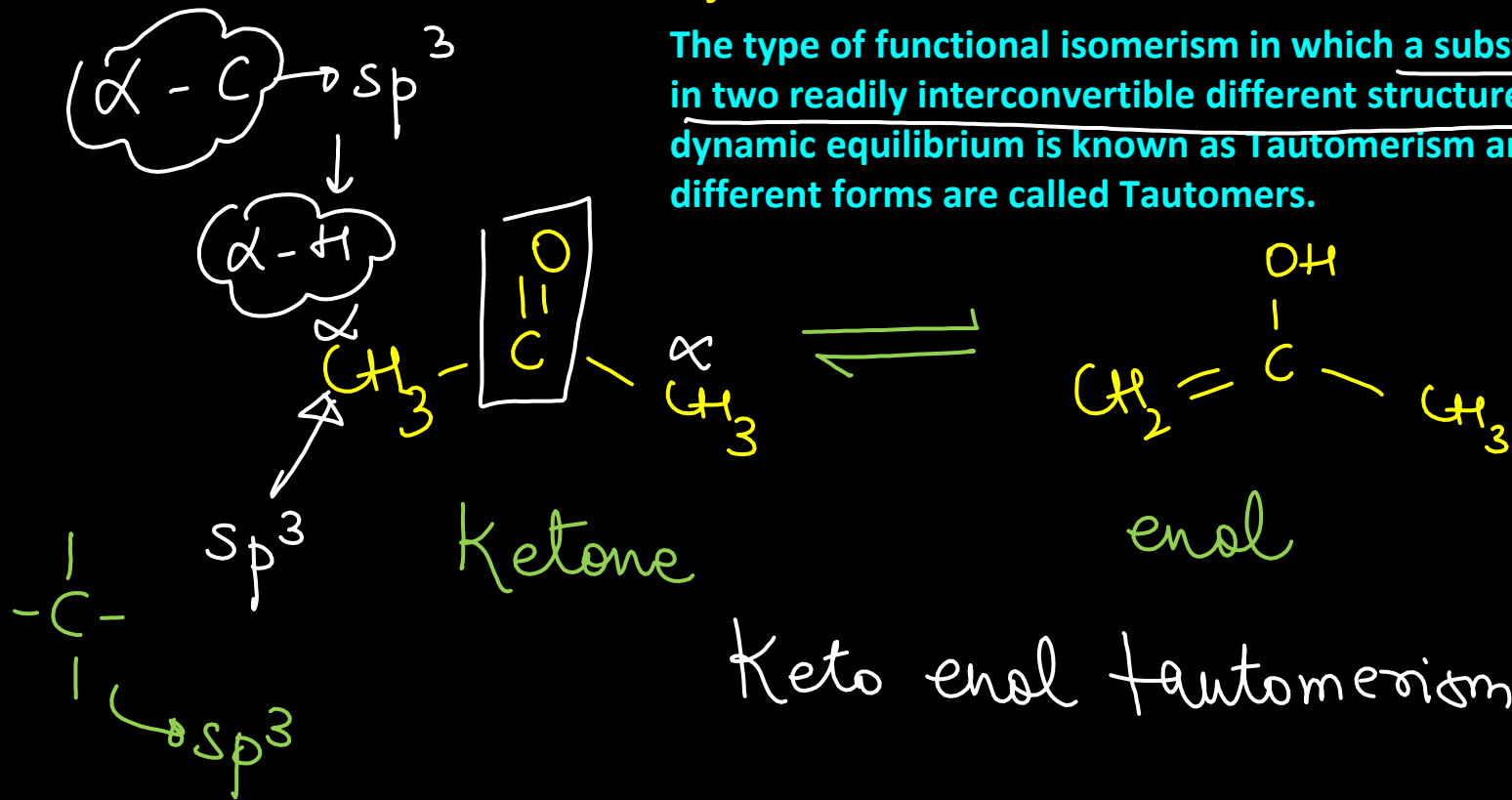
7



Tautomerism

↳ functional isomerism

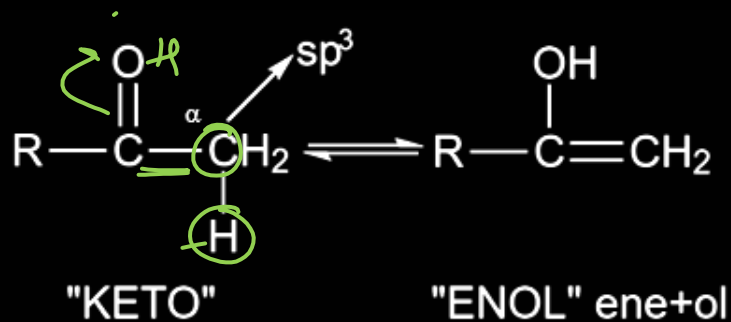
The type of functional isomerism in which a substance exists in two readily interconvertible different structures leading to dynamic equilibrium is known as Tautomerism and the different forms are called Tautomers.



Tautomerism

Example:

Keto-enol Tautomerism.

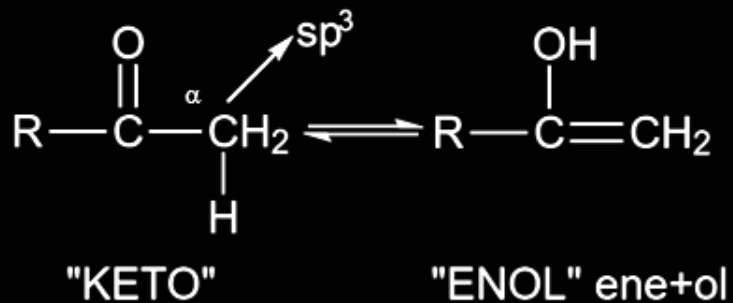


Acid-Base Theory:	W.A.	←	S.A.
Thermodynamics:	More stable		Less stable

Tautomerism

Example:

Keto-enol Tautomerism.



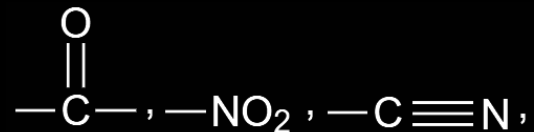
Acid-Base Theory:	W.A.	←	S.A.
Thermodynamics:	More stable		Less stable

Condition for Tautomerism :

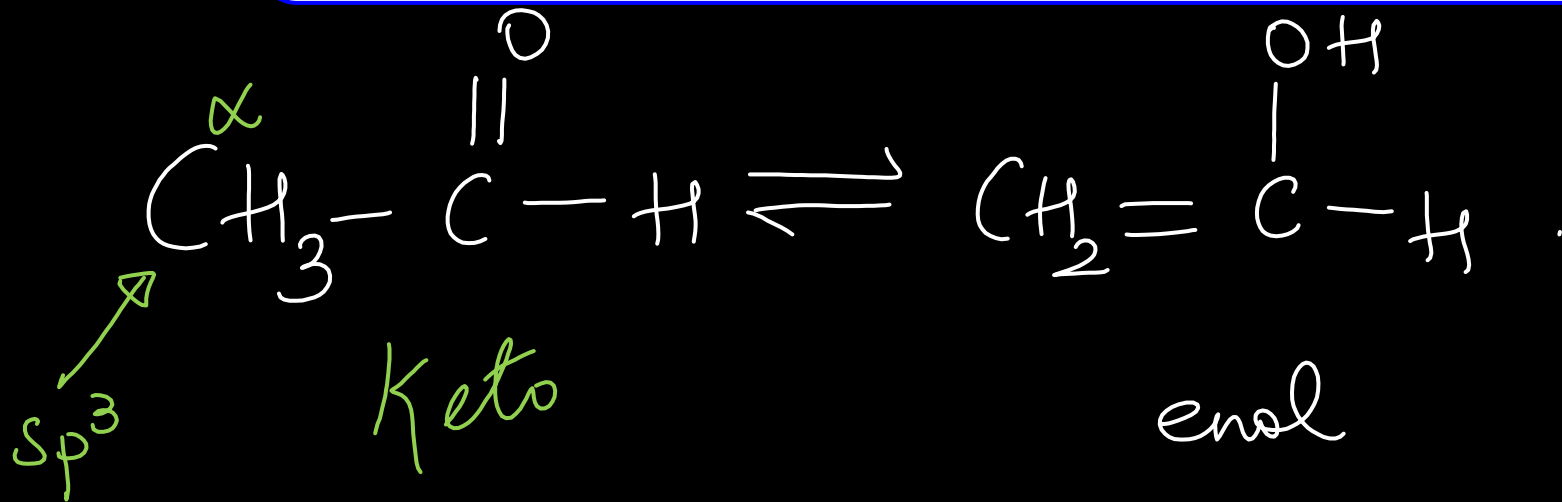
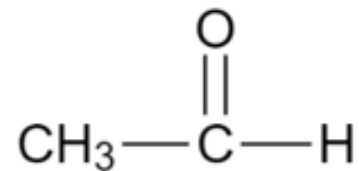
- ❑ Compound should have at least one α H, that is H attached to $\alpha - C$
- ❑ α H should be present on a sp^3 hybridised atom.

KETO	ENOL
C - H	O - H
C=O	C=C
C - C	C - O

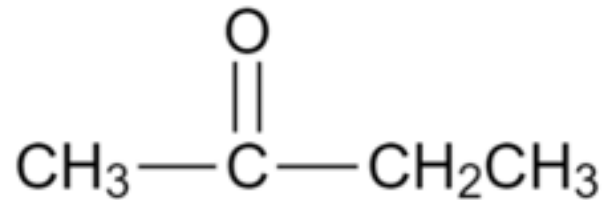
- ❑ Compound should have electron withdrawing



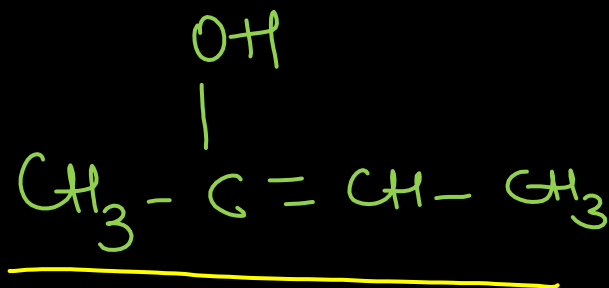
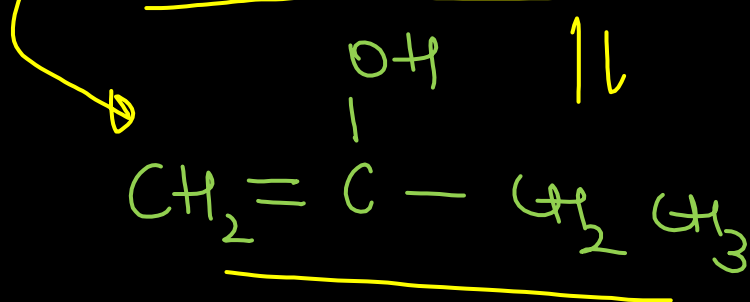
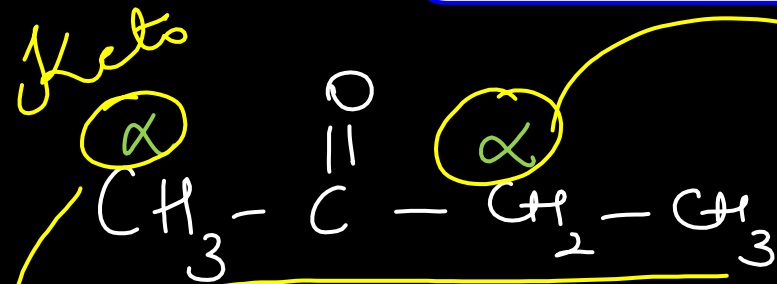
Draw the tautomers of:-



Draw the tautomers of:-

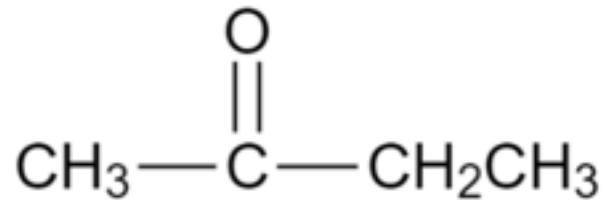


Keto

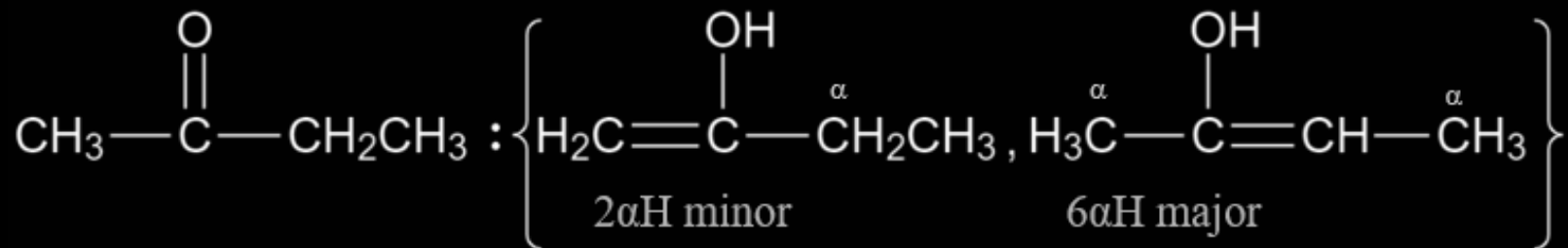


enol

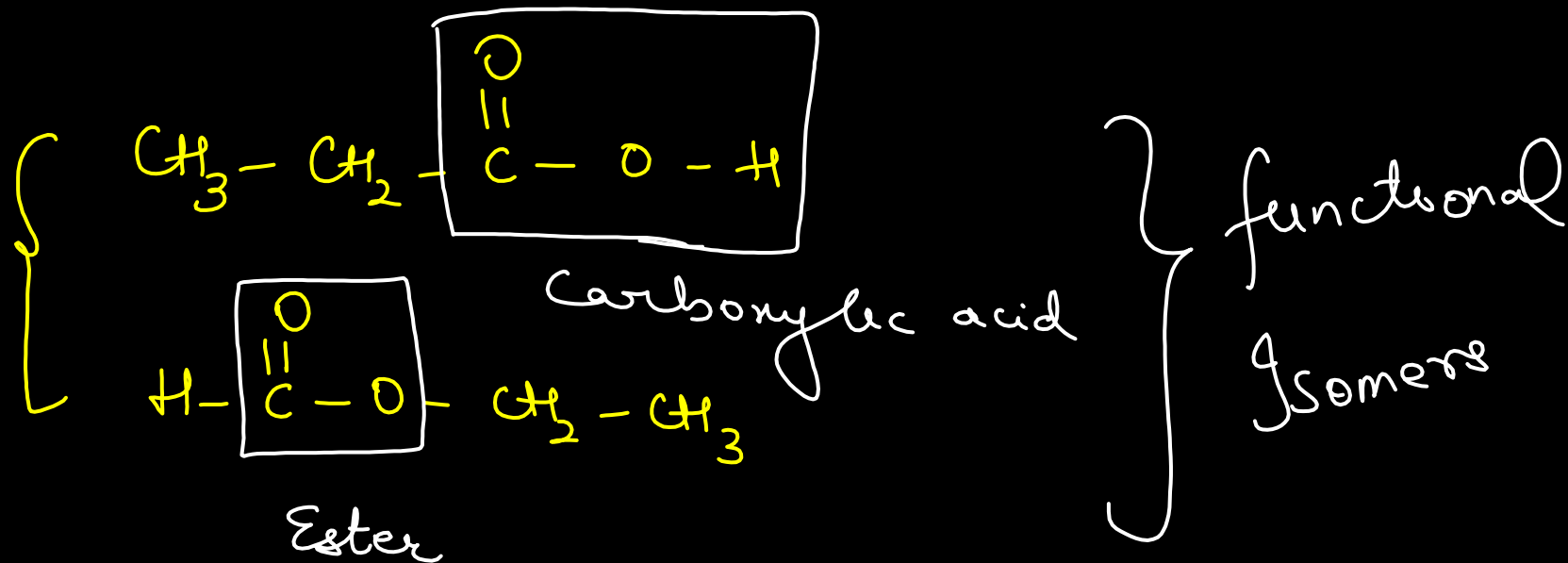
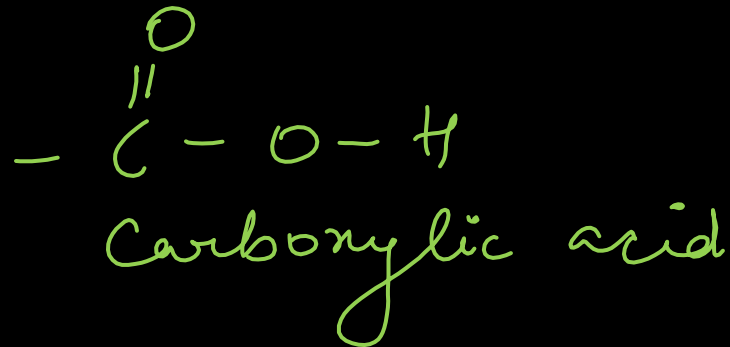
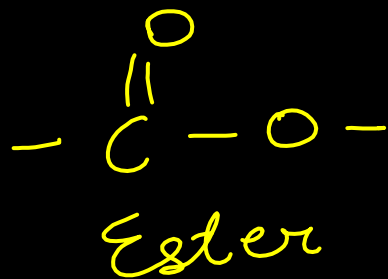
Draw the tautomers of:-



Solution



Stability of enol can be seen by stability of alkenes.



Stereo Isomerism

Isomers which have the same structural formula but have different relative arrangement of atoms or groups in space are called stereoisomers and the phenomenon is called stereoisomerism.

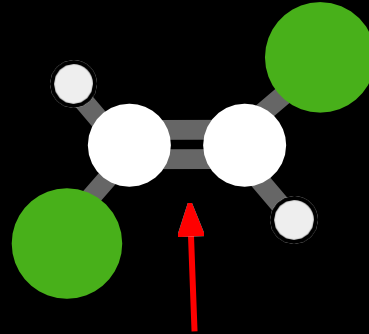
Geometrical isomerism

It arises due to restricted rotation of atoms or groups around a double bonded system or cyclic system

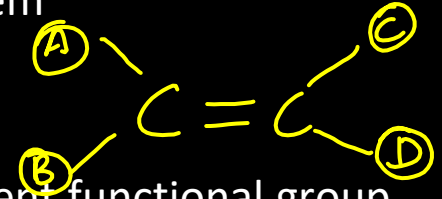
Conditions :

→ Restricted rotation

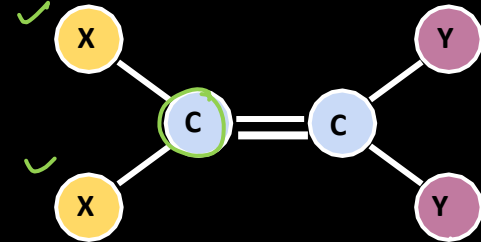
double
bond
or
Ring



No rotation about this double bond



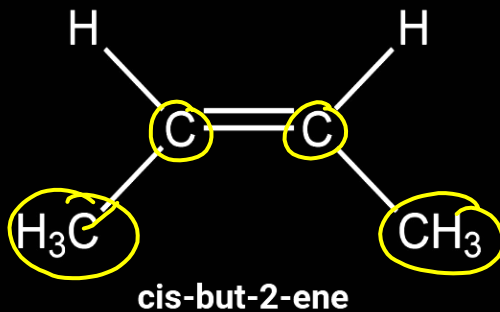
Different functional group
or atoms around system
showing GI



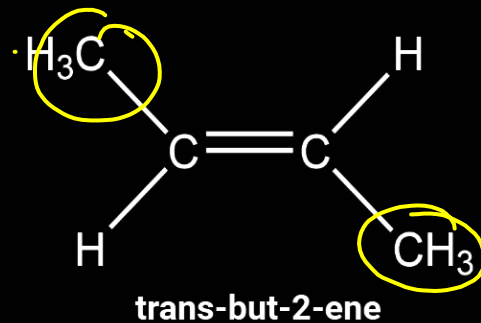
Not showing GI



Geometrical Isomerism



cis

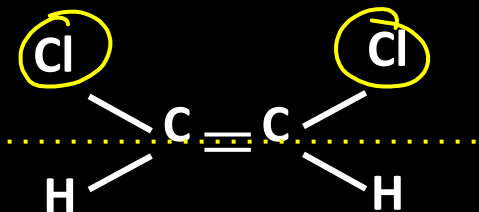


trans

The different arrangement of atoms in a space remains fixed due to **restricted rotation of double bond**

Geometrical Isomerism

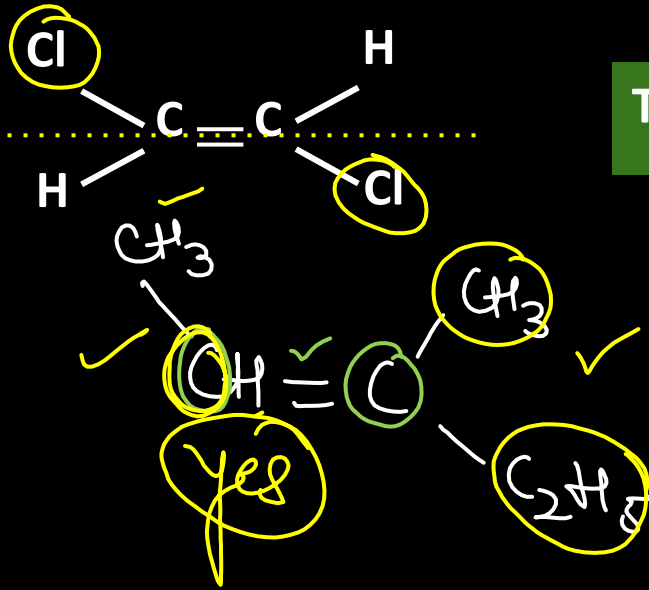
- When two Cl atoms (same atoms) are on the same side of the axis (or double bond) then it is called Cis – isomer



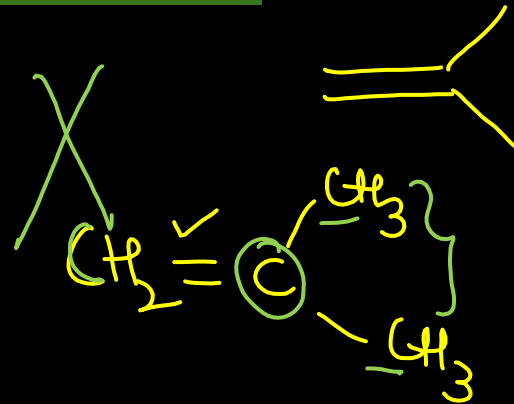
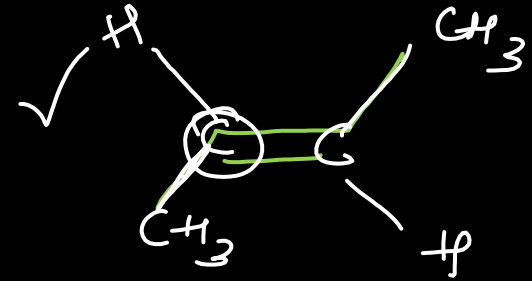
Cis isomer

Geometrical Isomerism

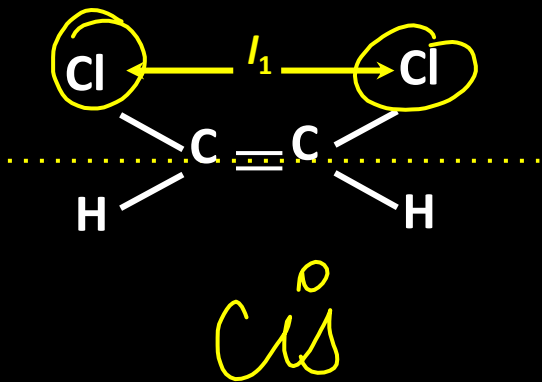
- When both Cl atoms are on opposite side of the axis (or double bond) then it is called Trans- isomer



Trans isomer



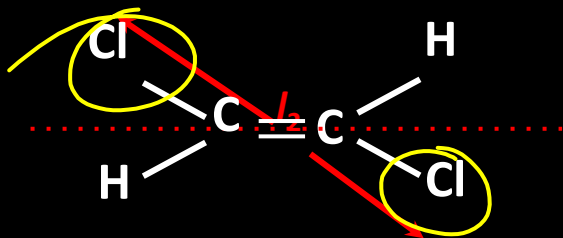
Geometrical Isomerism



Let the distance between two Cl atom be l_1 .
We define an axis along double bond.
We can see that two chlorine atoms are on same side to the axis.

Geometrical Isomerism

- Exchange the position of Cl and H atom of the right side carbon we get,

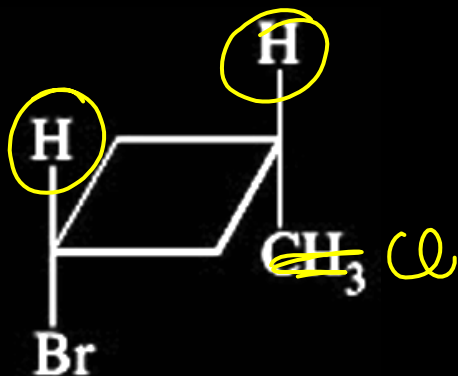


Observed :

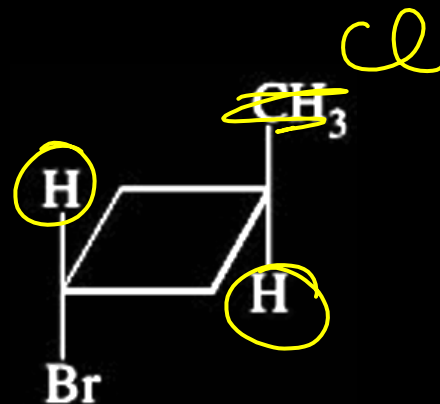
$$l_1 < l_2$$

Now we can see that Cl atoms are opposite to each other and the distance between them has changed.

Cis-trans isomerism in cyclic compounds

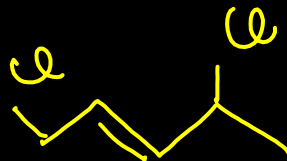
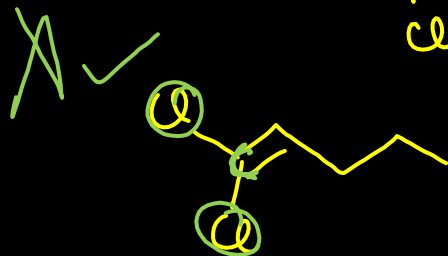
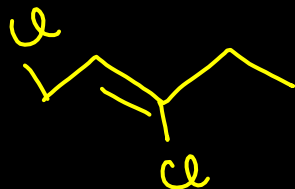
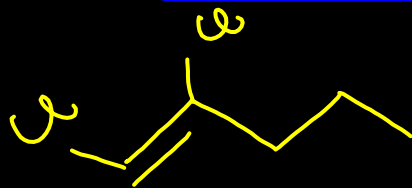


cis-1-bromo-3-chlorocyclobutane



trans-1-bromo-3-chlorocyclobutane

Which of the following does not show geometrical isomerism?



A 1, 2 Dichloro pent-1-ene

B 1, 3 Dichloro pent-2-ene

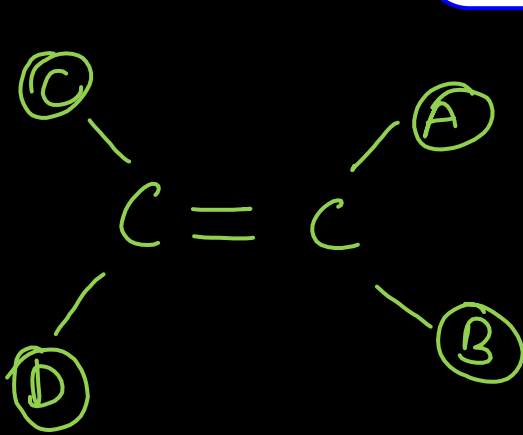
C 1, 1 Dichloro pent-1-ene

D 1, 4 Dichloro pent-2-ene

Which of the following does not show geometrical isomerism?

- A** 1, 2 Dichloro pent-1-ene
- B** 1, 3 Dichloro pent-2-ene
- C** 1, 1 Dichloro pent-1-ene
- D** 1, 4 Dichloro pent-2-ene

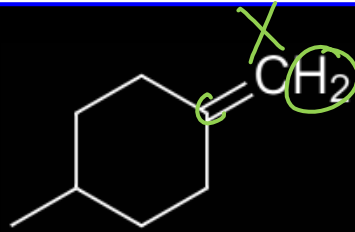
Geometrical isomerism is shown by



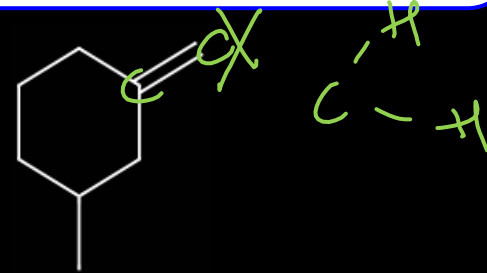
$A \neq B$

$C \neq D$

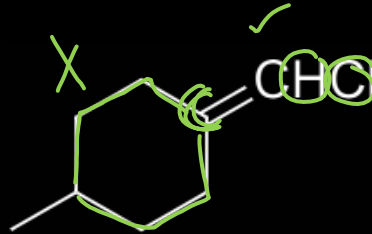
A



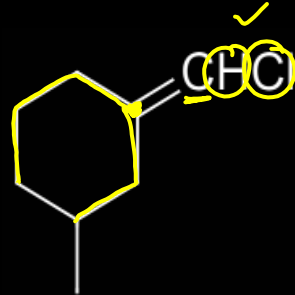
B



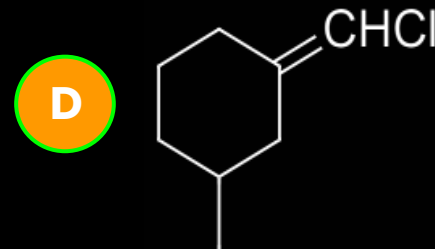
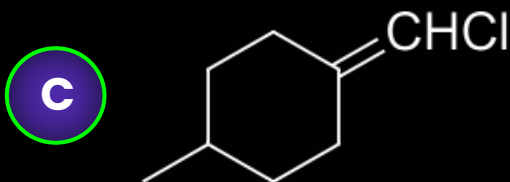
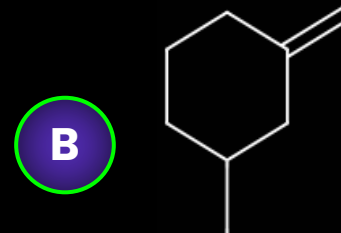
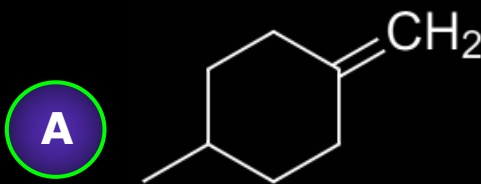
C



D

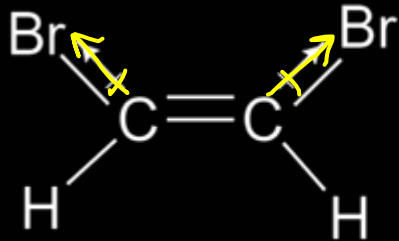


Geometrical isomerism is shown by



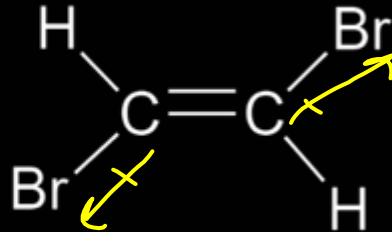
Properties of Cis- trans Isomers :

- ❖ Normally cis isomers have higher boiling point because of dipole interaction.



$\mu = 1.9 \text{ D}$

cis trans
more B.Pt



$\mu = 0, \text{ Non-polar}$

* more intermolecular
attractions



more B.Pt

Which of the following compound has highest boiling point and melting point respectively?

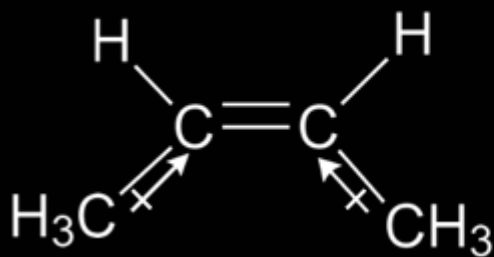
- A** **Cis-but-2-ene, cis-but-2-ene**
- B** **Trans but-2-ene, cis but-2-ene**
- C** **Cis but-2-ene, trans but-2-ene**
- D** **Trans-but-2-ene, trans but-2-ene**

Which of the following compound has highest boiling point and melting point respectively?

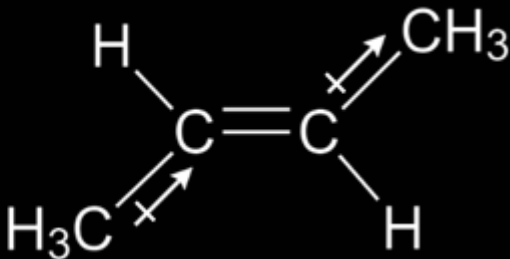
- A Cis-but-2-ene, cis-but-2-ene
- B Trans but-2-ene, cis but-2-ene
- C Cis but-2-ene, trans but-2-ene
- D Trans-but-2-ene, trans but-2-ene

Properties of Cis- trans Isomers

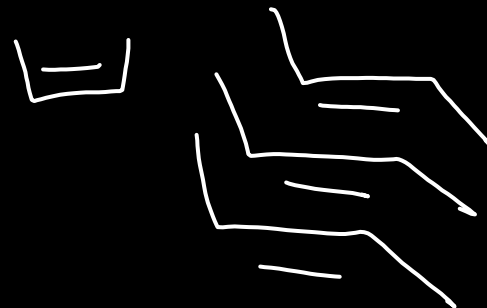
Cis isomer is polar having non zero dipole moment.



$\mu = 0.33 \text{ D}$
bp = 3.7°C
(Polar)

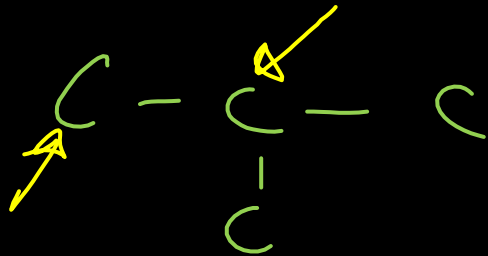
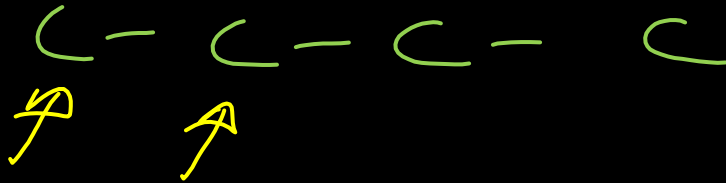


Bond dipoles cancel
bp = 0.9°C
(Non-polar)

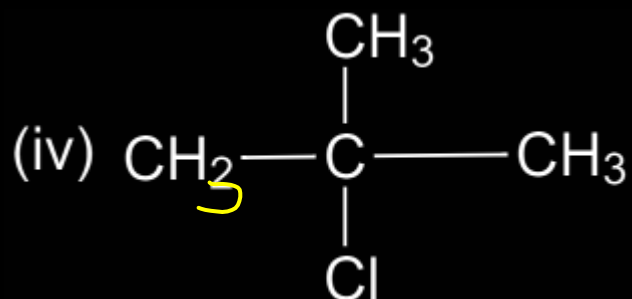
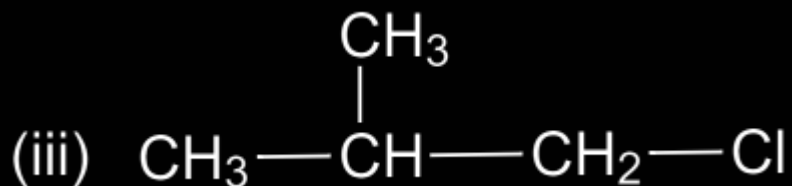
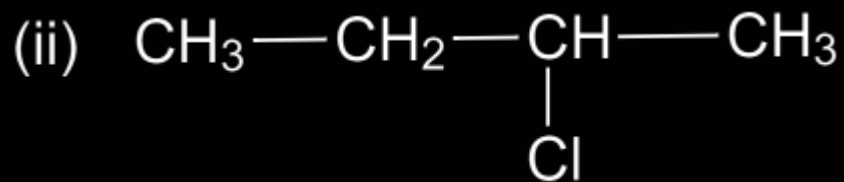


❖ Trans isomers have higher melting point because of better stacking (packing).

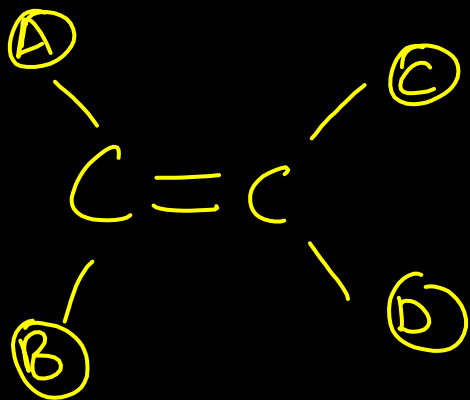
Draw all the possible isomers of monochloro Butane
(C_4H_9Cl)



Solution



The number of isomers for the compound with molecular formula $C_2BrClFI$ is



A 6

B 5

C 4

D 3

The number of isomers for the compound with molecular formula C_2BrClFI is

A

6

B

5

C

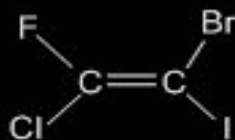
4

D

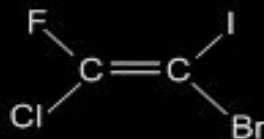
3

The number of isomers for the compound with molecular formula C_2BrClFI is

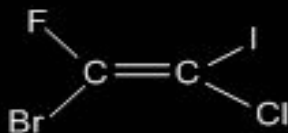
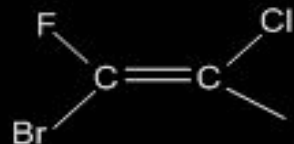
Solution



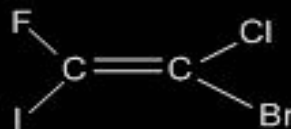
,



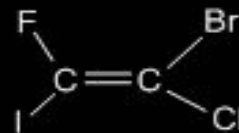
,



,

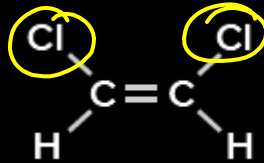


and

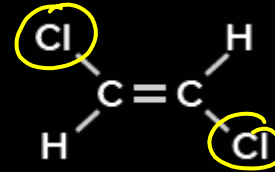


Cis and trans isomers

- **Cis Isomers** : Like atoms or groups attached at the same side of double bonded C-atom
- **Trans Isomers** : Like atoms or groups are on the opposite sides of doubly bonded carbon



Cis Isomer



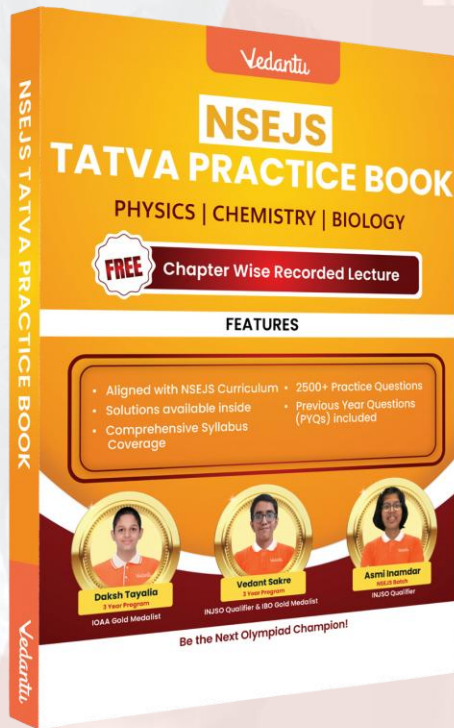
Trans Isomer

Properties	Value
Stability	Trans > cis
Boiling point	Cis > trans
Melting point	Trans > cis



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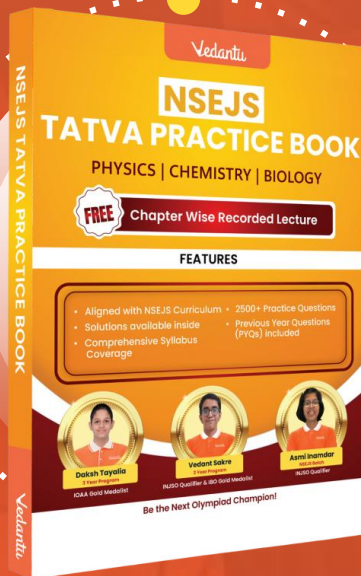


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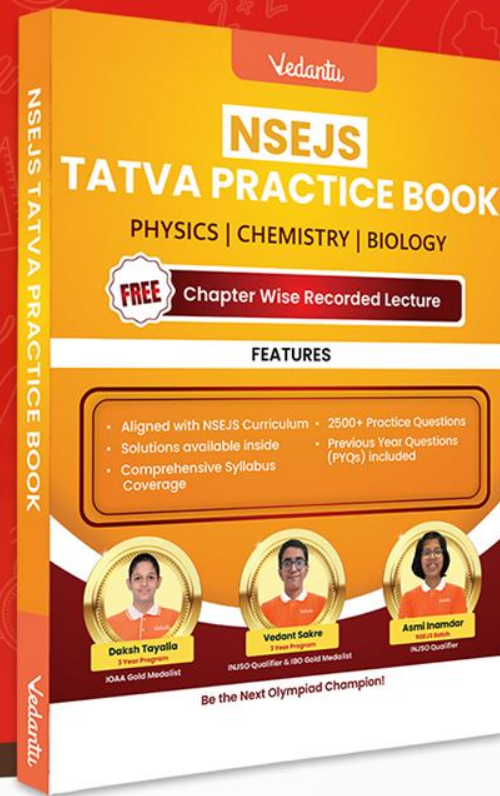
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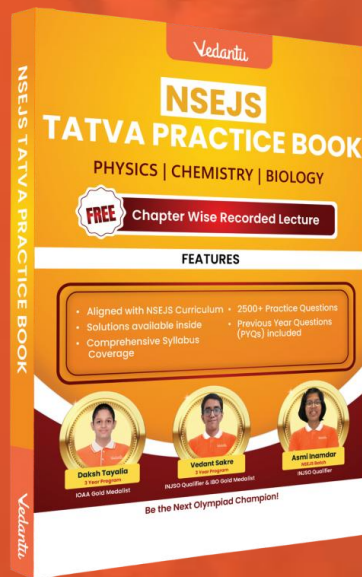
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